2 The matrices **A** and **B** are given by

$$\mathbf{A} = \begin{pmatrix} 1 & \frac{3}{2} \\ 0 & 1 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 1 & 0 \\ \frac{3}{2} & 1 \end{pmatrix}.$$

(a) Give full details of the geometrical transformation in the *x*-*y* plane represented by **A**. [2]

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(b) Give full details of the geometrical transformation in the *x*-*y* plane represented by **B**. [2]

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The triangle *DEF* in the *x*-*y* plane is transformed by **AB** onto triangle *PQR*.

(c) Show that the triangles *DEF* and *PQR* have the same area. [2]

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(d) Find the equations of the invariant lines, through the origin, of the transformation in the x-y plane represented by **AB**. [5]